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The palaeomagnetism of the Tertiary igneous complex of Ardnamurchan

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Summary. Samples have been taken systematically through the intrusive sequence for the three centres which form the Tertiary central intrusive ring complex. Palaeomagnetic measurements show that all the units are reversely magnetized as are the earlier lavas – an extension of those of Mull and Morvern – which they intrude. This is in contrast to Mull, Skye and Arran, where the igneous sequences have been shown to span at least three polarity intervals, but is similar to Rhum where the only normal units are a few late dykes. Since Rhum and Ardnamuchan both lie on the dyke swarm which passes through Muck, the two complexes are considered to be closely related. If the Ardnamurchan central complex was formed in a single reversed interval the total duration of igneous activity was less than 3 Myr.

Introduction

Ardnamurchan is the site of a classic, Tertiary central intrusive complex, made famous by the work of Richey & Thomas (1930). Excellent summaries of the geology have been given by Richey (1933), Deer (1969), Gribble, Durrance & Walsh (1976) and Emeleus (1982).

The aim of the work reported in this paper was to sample as completely as possible from the well-established igneous sequence (Table 1a) in order to determine the magneto-stratigraphy and to relate the Ardnamurchan central intrusive complex to the neighbouring ones of Mull (Mussett, Dagley & Skelhorn 1980), Rhum (Dagley & Mussett 1981) and Skye (Dagley & Mussett 1980).

The central intrusive complex has been intruded into Moine schists, and overlying thin sequence of Jurassic and Triassic sediments and Tertiary basaltic lavas which are considered to be an extension of the Mull–Morvern plateau lavas; the base of the lavas can be seen on the eastern flanks of Ben Hiant. Richey & Thomas (1930) recognized three successive centres of intrusion, Centre 1 to the east, Centre 2 to the west with Centre 3 between, and suggested a sequence of events on which Table 1 is based. However, there is some debate about the sequence, particularly within Centre 1.

Richey considered the Ben Hiant complex (events 2 and 3) to be early, and either between the plateau lavas and Centre 1 or part of the Centre 1 with possibly the youngest unit, the quartz-dolerite, contemporaneous with the Centre 1 cone-sheets (event 6). Gribble (1974, p. 76), as a result of a petrochemical investigation, has indeed suggested that the Ben Hiant quartz-dolerite represents several cone sheets which have coalesced within the earlier event. Field evidence shows that the quartz-dolerite cuts the porphyritic dolerite plug but there is no indication of the time interval.

An alternative interpretation for the Ben Hiant complex has been put forward by Green & Wright (1969, 1974). They argue that, assuming there has been negligible movement, the question arises as to how the subaerially-formed agglomerates, tuffs and pitchstone lavas of Ben Hiant can be preserved at a lower topographical level than the ostensibly younger but deeper-level intrusions of Centres 2 and 3, which are thought to have solidified beneath at least 1000 m of overburden. Thus they propose that the Ben Hiant sequence results from a flank eruption younger than either Centre 2 or 3, and that many of the so-called Centre 1 cone sheets were emplaced late in the development of the central intrusive complex. However, we believe that the Ben Hiant sequence is part of Centre 1 and field observations suggest that the inter-bedded agglomerates and tuffs, along with the andesitic pitchstone lavas which occur in the upper part of the bedded pyroclastic sequence, have been downfaulted into a small caldera, termed by Richey & Thomas (1930) the South-West Vent. This sequence of agglomerates, tuffs and pitchstone lavas is younger than the plateau lavas and may represent the surface manifestation of the Northern Vents.

There is one minor difference in our interpretation of the Ben Hiant sequence compared with that of Richey & Thomas (1930). We believe that the porphyritic dolerite plug post-dates the agglomerates, tuffs and pitchstone lavas whereas Richey & Thomas (1930, p. 151), on the basis of devitrification textures in the adjacent pitchstones, thought that the plug post-dated the lowest two pitchstones but pre-dated pitchstones 3 and 4 of this study. In a stream section (540 626) the porphyritic dolerite plug is seen to have an intrusive, chilled margin against tuffs and agglomerates which overlie the lowest two pitchstone lavas. Although no contacts are exposed between the porphyritic dolerite and the pitchstone lavas 3 and 4 the scarps formed by these pitchstones are terminated to the SW by the margin of the porphyritic dolerite.

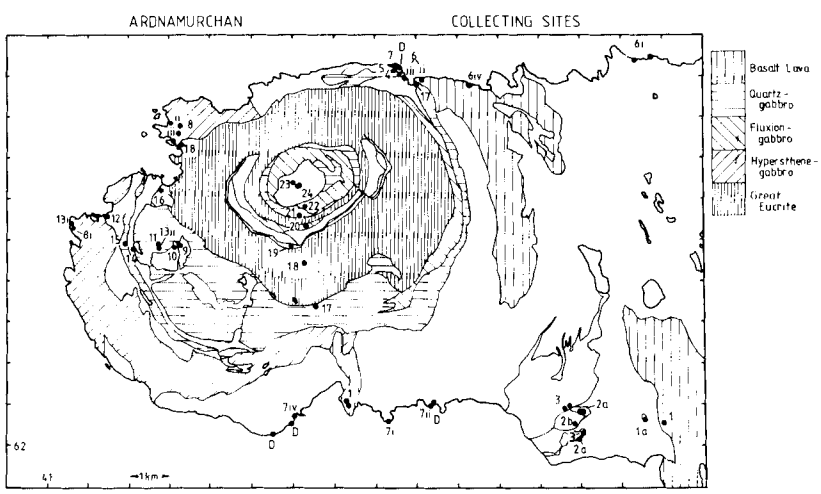


Figure 1. A simplified geological map of Ardnamurchan to show the sampling sites. The numbers refer to the events of Table 1 and D to dykes.

The trachyte plug (event 1a) is difficult to place precisely in the sequence; although it is intruded into Moine rocks and must be early because it is cut by Centre 1 cone-sheets.

The only previous palaeomagnetic work had been limited to specific problems. Khan (1960) sampled the Hypersthene Gabbro for comparison with gabbro bodies in Rhum and Skye. He found it to be reversely magnetized. Wells & McRae (1969) undertook four radial traverses across the Hypersthene Gabbro intrusion to ascertain whether the dips of the layering were a primary feature or produced by post-consolidation deformation. They concluded that because there was a close clustering of the palaeomagnetic directions about that found by Khan the layering had suffered little or no deformation after cooling below the Curie temperature of about 570°C.

Watt (1966) sampled some of the cone sheets to investigate the correlations of the magnetic properties with the fabric. His samples were reversely magnetized.

Intrusions from the three centres have been dated by the K-Ar method (see discussion for details) and the dates obtained suggest an age about 60 Ma, i.e. the same as most of the rest of the British Tertiary Igneous Province (Mussett 1980).

Sampling and results

Sampling was carried out using portable drills, orienting the cores by means of sunsights or geographic bearings. The 73 sampling sites are shown in Fig. 1 and listed according to the established sequence in Table 1(a). Only dykes which have a cross-cutting relationship with intrusions belonging to the central intrusive complex were sampled (Table 1b). The directions of magnetization of at least four cores from each site were determined using a parastatic magnetometer and incremental alternating field demagnetization. Thermal demagnetization was also used for samples from most units.

For some samples the magnetization was so strong that it is almost certain that the locality had been struck by lightning. In a few of these cases demagnetization uncovered a primary direction of magnetization on which the directions for several individual cores converged; in other cases no agreement was achieved.

In general, the results after thermal demagnetization are in good agreement with those following magnetic cleaning. However, frequently the change of direction was less than occurred during magnetic cleaning and in several cases there was no change from the NRM direction so that the primary direction is not revealed. Those units with noticeably different directions after demagnetization by the two methods are V005, V009B, V013, V020, V022, V043, V048, V060, V062 and V068.

The detailed results are listed in Table 2 (see also Figs 2 and 3). Several features of the results are worth pointing out.

The acid and basic components of each of the composite bodies sampled (V2 in Fig. 3c, V9 in Fig. 3b and V44 in Table 2) are significantly different. It is not uncommon for the acid parts to give more uncertain directions because of their weaker magnetization, as with V009 but for V002 it is the basic part which has the less well defined direction.

Similar differences occur between core clusters (sites) from different parts of the same body; thus V016 and V017 from the Quartz-gabbro of Faskadale (event 17) and V018 and V047 from the Great Eucrite (event 18) do not agree well. In these cases it may be naive to regard these bodies as simple intrusions. The Quartz-gabbro of Faskadale varies from a basic variety in the east to a more acid variety in the west (Richey & Thomas 1930). At Faskadale, the rock (V016) is an olivine-gabbro whereas the rock sampled (V017) to the east of the Achnaha-Kilchoan road is a quartz-gabbro with well-developed interstitial granophyric intergrowth. Because of the presence of internal contacts, combined with considerable petro-

Table 1. (a) Polarities for Ardnamurchan igneous complex in stratigraphic order.

Event No.	Unit		Code	Grid. Ref.	Polarity
Centre 3	24. Monzonite		V23	472 683	R
	23. Tonalite		V24	470 683	R
	22. Fluxion gabbro of Glendrian		V22	474 678	R
	21. Qtz-gabbro		V21	472 676	R
	20. Fluxion gabbro of Sithean Mor		V20	473 674	R
	19. Qtz-dolerite veined with granophyre		V19	471 669	R
	18. Great Eucrite		V018	473 665	R
			V047	443 694	
	Cone sheets				
	17. Qtz-gabbro of Faskadale		V016	500 708	R
			V017	477 662	
Centre 2	16. Fluxion gabbro of Portuairk		V45	437 687	R
	Younger qtz-gabbro of Beinn Bhuidhe		not sampled		
	15. Qtz-gabbro of Loch Caorach		V43	429 669	R
	14. Eucrite of Beinn nan Ord		V42		R
	13. Cone Sheets, Inner Set	i	V5,V8/V6	416 674/416 673	R
		ii	V40	437 668	
	12. Granophyric qtz-dolerite of Sgurr nam Meann		$V44^{A}_{B}$	424 675	R
	Qtz-gabbro of Aodainn		not sampled		
	11. Older qtz-gabbro of Beinn Bhuidhe		V41	437 669	R
	10. Grigadale Granophyre		V39	442 669	R
	Qtz-gabbro of Garbh-dhail		not sampled		
	9. Older gabbro of Lochan an Aodainn		V38/V37	443 668/443 670	R
	8. Hypersthene gabbro	i	V7	415 674	R
		ii	V25/V26	443 697/440 701	
		iii	V46/V55	443 696/412 694	
	7. Cone Sheets, Outer Set	i	$V1/V2^{A}_{B}$	494 627/494 626	R
			V3/V4	493 626/492 626	
		ii	V27	503 631	
		iii	V32	495 712	
		iv	V49/V58	471 628/470 627	
Centre 1	6. Cone Sheets	i	$V9^{A}_{B}$/V10	557 714/555 714	R
			V11/V12	554 713/553 714	
		ii	V13,V14/V15	500 709/501 709	
		iii	V29	498 709	
		iv	V61	510 707	
	5. Granophyre of Faskadale		V31	495 712	R
	4. Quartz-gabbro of Faskadale		V30	497 710	R
	3. Quartz-dolerite of Ben Hiant (could belong to event 6)	Ben Hiant complex	V61/V69,V70	540 624/538 631	R
	2b. Porphyritic dolerite plug		V52/V65,V66	538 625/540 626	N/R
	2a. 4th Andesitic pitchstone lava		V68	541 628	R
	3rd Andesitic pitchstone lava		V67	542 627	R
	1st Andesitic pitchstone lava		V50	539 623	R
	Bedded tuffs and agglomerates				
	1a. Trachyte plug		V62	556 626	R
	1. Plateau basalt lavas		V34,V35,V36	561 625	R
			V63/V64	483 632/484 630	

Table 1 (b)

Dykes sampled	Code	Grid ref.	Polarity
Cut by cone sheet V27 (Outer Set of Centre 2), Mingary Castle	V28	504 631	R
Basic dyke cutting cone sheet V27	V56	503 631	R
" " " " "	V57	503 631	R
Sron Bheag dolerite dyke. Cut by Outer Set of Centre 2 cone sheets	V59	466 623	R
Sron Bheag acid dyke. Cut by Outer Set of Centre 2 cone sheets	V60	465 623	N
Cuts Centre 2(?) cone sheet and Faskadale Granophyre. W. side of Faskadale Bay	V33	495 712	R
Acid pitchstone dyke cutting cone sheet V49 (Outer Set, Centre 2), E. of Sron Bheag	V48	471 628	R
Basic dyke cutting cone sheet V49 (Outer Set, Centre 2), Sron Bheag	V53	471 628	R

logical variation, Richey & Thomas (1930) proposed that the intrusion may be multiple in origin. This might account for the magnetic differences between the rocks sampled at the two sites.

Sample V018 of the Great Eucrite is a plagioclase-pyrxoene-olivine cumulate obtained from a locality in the central part of the intrusion where indistinct layering is developed.

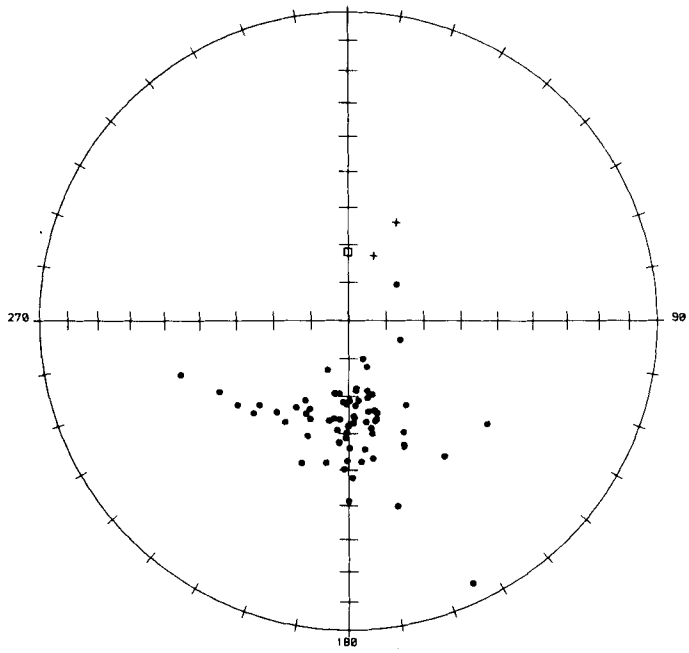


Figure 2. Palaeomagnetic site mean directions: equal-area stereographic projection. • negative inclinations, + positive inclinations, □ indicates the direction of the present dipole field.

V047 is an olivine gabbro from the margin of the intrusion. Deer (1969) has proposed that the Great Eucrite is not a ring dyke but a funnel-shaped (inverted cone) intrusion with crystallization taking place from the margin towards the centre of the intrusion. Presumably, it could be argued that the magnetic difference between the two samples, V018 and V047, is due to the different times of cooling to the blocking temperature in this, the largest intrusion of Centre 3.

The Hypersthene Gabbro exhibits confocally dipping layering which steepen inwards, from less than 10° to over 50°, towards a central area occupied by later intrusions. Skelhorn & Elwell (1971) believed that there was convincing evidence that the layering was produced by bottom accumulation of crystals on successive floors which were more or less horizontal so that the inward steepening must be due to later deformation produced by movement on a ring fault. In the present study, although no attempt was made to duplicate the work of Wells & McRae (1969), the Hypersthene Gabbro was sampled at five localities which included the quartz-gabbro marginal facies (V007, dip ~ 10°E) at the lighthouse and four sites at Sanna where the dip of the layering sampled is 15° (V026), 23° (V025), 30° (V046) and *c.* 33° (V055) in an approximately southerly direction. The site-mean directions for these (Fig. 3a), though they are not identical and are generally steeper than Khan's (1960) result ($D = 184.1°$, $I = -54.5°$), cluster reasonably close together. If the layers are returned to the horizontal the dispersion increases indicating that there has been no deformation of the layering below the Curie point, in agreement with the earlier systematic study.

Samples of the Ben Hiant quartz-dolerite are also discordant for V069 and V070 give directions somewhat different from V051. The samples V069 and V070 were collected close to each other within the main outcrop whereas V051 was obtained from an outlier 500 m to the south and at a lower topographic level. Texturally, V051 differs from V069 and V070 in having large areas of mesostasis consisting of skeletal plagioclase in a devitrified matrix. It is

Table 2. Palaeomagnetic results.

		Intensity			In-plane						
Code	N	D	I	κ_{95}	λ	Φ	J_0	dm	Polarity	T_c	Comments
A03V001	8	155	-57	9.8	-64	48	0.8	1.1	R	520	
A03V002A	8	145	-63	5.3	-65	69	6.5	8.3	R	542	Basic } parts of same sheet
A03V002B	8	164	-65	2.4	-76	48	3.2	4.0	R	537	Acid }
A03V003	8	165	-64	1.3	-75	38	1.6	2.0	R	525	
A03V004	8	162	-64	1.7	-73	40	2.0	2.6	R	530	
A03V005	8	157	-59	1.5	-68	51	1.4	2.1	R	540	
A03V006	8	174	-63	2.1	-75	35	2.3	2.9	R	540	
A03V007	8	181	-59	1.6	-71	358	1.8	2.5	R	550	c.f. 25, 26, 46, 55
A03V008	8	141	-51	3.6	-41	289	3.2	4.8	I	562	high intensities, lightning possible, though fairly good agreement
A03V009A	8	161	-53	1.7	-48	3	1.5	2.2	N	580	Basic } parts of same sheet; see dk
A03V009B	8	166	-61	3.0	-72	45	4.1	5.8	N	580	Acid }
A03V010	8	160	-41	3.1	-57	64	2.3	4.3	N	480-540	
A03V011	8	153	-73	7.0	-56	94	6.7	9.6	N	510-540	
A03V012	8	173	-69	2.6	-84	46	2.8	3.8	R	560	
A03V013	8	54	-74	10.5	-34	145	17.2	19.0	I	557	High intensities and directional spread, lightning
A03V014	6	156	-7	8.2	-43	25	4.1	8.3	I	560	Some high intensities, lightning; core 01 has usual direction
A03V015	8	165	-65	1.8	-75	40	1.5	2.3	R	567	
A03V016	10	137	-43	3.0	-41	67	3.4	4.7	I	560	normal intensities and well grouped
A03V017	8	161	-64	1.9	-76	355	1.1	1.4	R	567	same unit
A03V018	8	173	-55	1.0	-69	11	1.0	1.5	R	567	see 47
A03V019	8	204	-65	1.1	-72	289	1.5	1.8	R	560	
A03V020	4	110	-75	3.6	-56	123	6.1	6.7	I*	520-575	*Reversed on basis of geographic pole: some high intensities and scatter, lightning?
A03V021	10	200	-65	2.4	-70	280	9.6	11.8	R	567	
A03V022	10	177	-62	2.1	-67	98	8.7	11.2	R	560-580	Core directions converge; high intensities, lightning effect not fully removed
A03V023	6	200	-61	3.4	-75	338	3.9	5.1	R	568	
A03V024	8	185	-64	0.7	-78	337	0.8	1.1	R	552	
A03V025	8	165	-71	3.4	-81	81	5.1	5.9	R	562	c.f. 7, 26, 46, 55
A03V026	8	176	-64	0.7	-79	7	0.9	1.1	R	520-560	c.f. 7, 25, 46, 55
A03V027	8	180	-62	2.7	-75	306	2.5	4.6	R	560	cuts 28
A03V028	8	188	-60	1.8	-81	120	1.6	2.4	R	570	cut by 27
A03V029	8	156	-53	1.7	-62	40	1.6	2.3	R	550	
A03V030	8	252	-42	4.8	-30	264	3.7	6.0	I	565	normal intensities, but scatter
A03V031	8	226	-55	2.2	-53	279	2.2	3.1	R	529	cut by 33
A03V032	8	201	-62	2.4	-71	300	2.9	3.7	R	565	
A03V033	8	200	-66	4.6	-71	278	4.9	6.0	R	565	cuts 31
A03V034	6	196	-70	1.4	-84	280	2.1	2.4	R	470-580	
A03V035	8	170	-63	2.7	-76	25	3	3.4	R	480-540	
A03V036	8	165	-38	3.0	-53	18	2.1	3.5	R	565	cores converge but primary direction perhaps not reached
A03V037	6	167	-65	4.1	-78	39	5.3	6.6	R	525-555	
A03V038	8	177	-63	2.0	-77	4	2.8	3.5	R	566	
A03V039	8	166	-69	1.4	-81	67	2.0	2.4	R	567	
A03V040	8	175	-68	0.8	-83	19	1.1	1.3	R	560	
A03V041	8	175	-69	2.3	-85	29	3.4	4.0	R	547	
A03V042	8	187	-71	1.3	-86	287	1.9	2.2	R	555-585	
A03V043	8	188	-64	3.9	-77	327	4.9	6.2	R	575	
A03V044A	8	184	-58	1.3	-71	343	1.4	2.0	R	540-565	Basic } parts of same sheet
A03V044B	8	181	-60	2.9	-75	352	3.3	4.3	R	545-560	Acid }
A03V045	8	163	-60	2.9	-75	47	3.7	4.6	R	537	
A03V046	8	173	-72	5.6	-86	79	8.6	9.9	R	555-575	c.f. 7, 25, 26, 55
A03V047	8	177	-65	0.9	-80	7	1.1	1.4	R	560	see 18
A03V048	8	175	-52	4.0	-68	5	3.8	5.5	R	510	cuts 49
A03V049	8	158	-52	0.7	-76	131	1.1	1.4	R	567	cut by 48
A03V050	8	241	-58	2.5	-63	280	2.7	3.7	R	440/565	
A03V051	8	192	-54	3.1	-67	41	1.1	1.5	R	unclear	c.f. 69, 71, - 1 core different ? lightning

Table 2 – *continued*

		direction			pole position						
Code	N	D	I	α_{95}	λ	φ	dp	dm	Polarity	T_c	Comments
A03V052	10	26	61	2.4	68	114	2.8	3.7	N	520	c.f. 65, 66 - cores agree well
A03V053	8	159	-79	0.8	-74	145	1.5	1.6	R	565	cuts 49
A03V054	8	103	-76	1.2	-77	224	2.0	2.1	R	567	cuts cone sheet OS Centre 2
A03V055	8	163	-64	4.9	-75	47	6.3	8	R	550	c.f. 7, 25, 26, 46
A03V056	8	168	-61	1.1	-73	27	1.3	1.8	R	555	cuts 27
A03V057	8	178	-63	0.9	-77	2	1.1	1.4	R	550	cuts 27
A03V058	8	183	-69	2.8	-86	328	1.2	1.4	R	557	
A03V059	8	179	-56	1.3	-60	355	1.4	1.9	R	570	cut by OS Centre 2 CS
A03V060	6	22	72	3.3	78	76	5.2	5.9	N	not defined	very weak; uncertain. Acid dyke cut by OS Centre 2 CS.
A03V061	6	211	-63	1.6	-67	283	2.0	2.5	R	552	
A03V062	8	178	-48	5.6	-62	357	4.7	7.2	R		
A03V063	8	168	-59	4.0	-72	25	4.5	6.0	R		
A03V064	8	218	-59	1.3	-60	282	1.4	1.9	R		
A03V065	8	181	-68	1.9	-84	346	2.7	3.2	R		c.f. 52, 66
A03V066	8	162	-70	1.0	-79	74	1.5	1.8	R		c.f. 52, 65
A03V067	8	200	-54	1.9	-64	314	1.9	2.7	R		
A03V068	8	205	-63	0.9	-75	292	1.1	1.4	R		
A03V069	8	227	-57	1.6	-56	276	1.7	2.4	R		c.f. 51, 70
A03V070	8	233	-53	5.3	-47	274	5.1	7.3	I		c.f. 51, 69
OVERALL MEAN (62 units excluding 7 intermediates (2 with α_{95}>5.0), 2 normals (V52 & 60) and 2 with α_{95}>5.0)											
	62	180	-63	2.7	-78	355	3.3	4.2	R		using mean pole as definition of intermediates (8, 13, 14, 16, 20, 33, 70)
					(-78	357)					using geographic pole (includes 20)
(cf. Rhum centre 177 -58 4.4) Dagley & Mussett, 1981											
TYPE MEANS:											
V002 (A & B)	2	154	-64	18.1	-71	62	23.2	29.9			
V009 (A & B)	2	180	-67	21.0	-71	354	22.4	30.7			
V044 (A & B)	2	183	-59	7.3	-73	347	8.11	10.8			
7, 25, 26, 46, 55	5	172	-66	5.9	-81	27	7.9	9.7			Hypersthene gabbro
(cf. Rhum	6	184	-55)								
18 & 47	2	174	-60	20.8	-74	10	23.8	31.4			Great Eucrite
51 & 69 & 70	3	206	-60	17.9	-67	296	43.1	57.2			Quartz dolerite of Ben Hiant
69 & 70	2	230	-55	12.6	-51	275	12.7	17.9			
65 & 66	3	172	-69	15.8	-84	46	22.8	26.9			Porphyritic dolerite of Ben Hiant

N = no. of directions used to find mean; $N/2$ = no. of independently oriented cores; D = declination, east of true north; I = inclination, positive downward; α_{95} = semi-angle of cone of 95 per cent confidence; λ = latitude of palaeomagnetic pole, positive north; ϕ = longitude of palaeomagnetic pole, east of Greenwich; *dp*, *dm* = semi-axes of oval of 95 per cent confidence. Polarity: N = normal ($0 < \theta < 40°$ where θ is the polar colatitude); I = intermediate ($40 < \theta < 140°$); R = reversed ($140 < \theta < 180°$). Tc = Curie point temperature (°C).

not certain whether this difference indicates two separate intrusions or just one, with the textural differences due to proximity to a contact.

The most striking disagreement is between the samples of the porphyritic dolerite plug of Ben Hiant. V052 gave a normally magnetized direction but V065 and V066 which were collected at a different locality, 100 m to the NE to check this surprising result both gave reverse directions of magnetization which agree well. Demagnetization of the NRM of V052 and of anhysteretic, isothermal and thermal remanences induced in it have failed to reveal any reason why it should record a normal direction unless the field direction when it cooled was normal. However, this seems an unlikely possibility. Thermal overprinting of V052

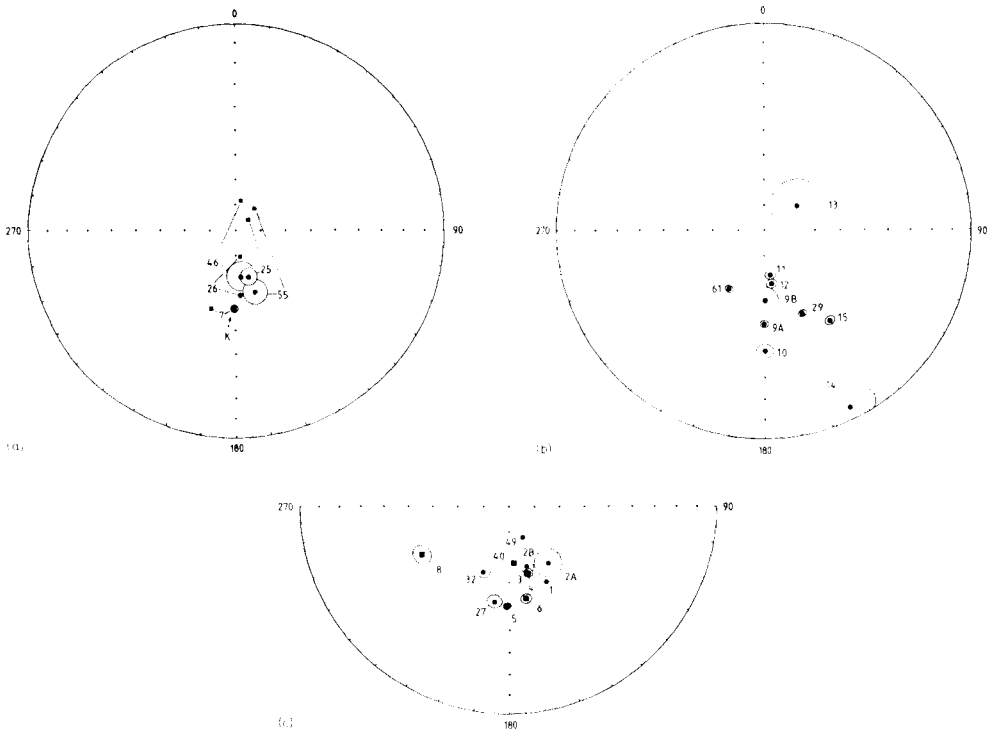


Figure 3. Palaeomagnetic site mean directions: equal area stereographic projections. Numbers refer to code given in Table 2. Where included, ovals indicate 95 per cent of confidence limits, otherwise they are too small to show. (a) Hypersthene Gabbro; ● directions listed in Table 2, ■ directions when layering is made horizontal, K, ♦ Khan's result (1960). (b) Cone Sheets of Centre 1. (c) Cone Sheets of Centre 2: ● outer set, ■ inner set.

cannot be ruled out, though, in contrast to those sampled, the body concerned would have to be normally magnetized. A lightning strike is also a conceivable explanation as some samples have moderately strong magnetizations ($\sim 2 \times 10^{-3}$ A m^2 kg^{-1}).

The only other body with apparent normal magnetisation is the Sron Bheag feldspar-porphyritic felsite dyke V060. This dyke has been extensively altered, with the development of calcite and secondary white mica, the pyroxene has been pseudomorphed and the meagre amounts of iron oxide occur in areas where there is much calcite. The samples were very weakly magnetized so that it was not possible to examine the remanence more thoroughly and we suspect that it is viscous remanence acquired in the present field.

The cone sheets belonging to Centre 1 (V009, 010, 011, 012, 013, 014, 015, 029, 061), which were sampled at four geographically separated locations (Fig. 1), do not form a tight group of palaeomagnetic directions (Fig. 3b). Group 6i (V009, 010, 011, 012) indicate two directions defined by V009A and 10 and V011 and 012. Two (V013, 014) of the three collected at location 6ii were strongly magnetized (up to 10^{-1} A m^2 kg^{-1} for V013 and 7×10^{-2} A m^2 kg^{-1} for V014) but give somewhat scattered results about anomalous directions and it is probable that these have been affected by lightning. The same is true for V061 (6iv) (up to 2×10^{-2} A m^2 kg^{-1}), from which some samples had to be rejected.

The differences in the directions between V009A, 010, 011, 012, 015 and 029 may arise because the conesheets were emplaced over a period of time during which the magnetic field had changed through secular variation.

Similarly for the Inner Set of Cone Sheets of Centre 2 (Fig. 3c). Although magnetic cleaning of V008 yields fairly well-defined directions the strong magnetization of some cores suggests that they may have been affected by lightning. The demagnetization of V040, which is located closer to the focus of Centre 2 and shows signs of having been metamorphosed by Centre 3, gives a well-defined direction but also reveals a secondary normal component of magnetization. A similar spread is seen in the directions for the Outer Set of Cone Sheets of Centre 2 (V001, 002, 003, 004, 027, 032, 049 and 058) (Fig. 3c) though the agreement amongst the four cone sheets (001–4) sampled just to the west of Mingary Pier is quite good.

Of the 73 sites, two gave normal, 64 reversed and seven intermediate directions, defined in terms of the criteria of Wilson, Dagley & McCormack (1972) (N, $0 < \theta < 40°$; I, $40° < \theta < 140°$; R, $140° < \theta < 180°$), but using the mean palaeomagnetic pole rather than the geographic pole to define the axis of the ancient magnetic field (only V020 would be classified differently as R rather than I if the geographic axis had been used).

The overall mean direction (180°, −63°, $\alpha = 2.7°$) and pole position (−77°N, 355°E) were calculated using the 62 reversed site mean directions with $\alpha_{95} < 6.0°$; this excluded V021 and V022.

The thermomagnetic behaviour of samples from each site was determined using a strong field Curie balance. There was no evidence for 'double' Curie points and for some samples the Curie transition was very poorly defined. Curie points generally lie in the range 500–580°C (Table 2) suggesting titanomagnetite as the principal carrier of the remanence.

Discussion and conclusions

It is our contention that the primary polarity of all the bodies sampled is reversed and that the igneous sequence was formed during a single reversed interval of the geomagnetic field. However, before interpreting our results further the alternatives involving slow cooling and/or overprinting must be considered.

If, during the evolution of the three centres, the regional temperature never fell below the Curie point, the magnetic polarities will reflect only the post-Centre 3 cooling event. A reason for such slow cooling could be that, at the present level of erosion, the Ardnamurchan central intrusive complex represents a deeper level than for instance, the Mull complex. The Ardnamurchan complex is characterized by an abundance of basic intrusions with only one acid intrusion of any size, the Grigadale Granophyre measuring 0.5 km in diameter, whereas at the present level of erosion acid intrusions comprise 40 per cent of the Mull complex.

We consider this hypothesis of slow cooling unlikely because of the small size of the intrusive complex. From the interpretation of gravity anomalies, Bott & Tuson (1973) have shown that the Ardnamurchan complex is probably underlain by a cylinder of mafic material having a diamter of 12 km extending to a depth of 4.5 km beneath the present surface. This is much smaller than is found for the Mull and Skye complexes for both of which the underlying cylinders are estimated to be 20 km in diameter and 14 km in depth. We have estimated in the case of Mull (Mussett *et al.* 1980) that a cylinder 10 km in diameter would take no longer than about 0.44 Ma to cool from 1000 to 350°C which is less than the duration (*c.* 0.67 Ma) of normal polarity intervals at this time (Harland *et al.* 1982). Not only is it likely that the cooling of the Ardnamurchan complex took an even shorter time but also the critical interval is that between the Curie temperature (580–500°C) and the principal blocking temperature (in most cases higher than 350°C) so that, if any part was emplaced during a normal interval, that polarity should have been recorded.

On the other hand, if normal polarity had been recorded intially by some of the intru-

sions they may have been reheated and magnetically overprinted by later intrusions, particularly those of Centre 3, to give the common reversed polarity.

With this possibility in mind, intrusions belonging to both Centre 1 and 2 were sampled geographically as far away and as close as possible to Centre 3. For example, Centre 1 basic cone sheets were sampled to the east of Ockle Point, Port Ban and Faskadale, respectively 5, 1.2 and 240 m from the margin of Centre 3. Within Centre 2, a 2.5 km long traverse was carried out from the Point of Ardnamurchan (lighthouse), through Grigadale to Achosnich in which various major intrusions and cone sheets were sampled. We think that this geographical disposition of the sampling sites is greater than the likely width of the zone of reheating. In Mull, we have shown (Dagley, Mussett & Skelhorn 1983) that the zone of magnetic overprinting produced by the major intrusions of Centre 2 has a width of approximately 1 km.

Those intrusions in the Ardnamurchan Centre 2 traverse, the Older Quartz Gabbro of Beinn Bhuidhe (V041), a member of the Inner Set of cone sheets (V040), the Grigadale Granophyre (V039) and the Older Gabbro of Lochan an Aodainn (V037, V038), which are nearest to Centre 3 all exhibit evidence of metamorphism as shown by intense clouding of the plagioclase and the development of iron oxide in the ferromagnesian minerals, similar to that which in Mull is associated with magnetic overprinting. However, all the intrusions in this traverse give reversed polarity, and thermal demagnetization of cores from V039, V040 and V041 shows that a secondary normal component is removed by 300°C. There is similar, but weak, evidence for overprinting by normal magnetization in some of the later intrusions. Thus this evidence suggests that the reversed polarity of the unmetamorphosed intrusions of Centre 2 is primary.

We believe, therefore, that all the intrusions we have sampled, as well as the dykes and lavas, are reversely magnetized. The reversed polarity of the lavas is consistent with the belief that they are an extension of the Mull–Morvern lavas. Seven of the eight dykes sampled are reversed; the other V060 gave an unreliable result.

Though the Ardnamurchan complex consists of three consecutive centres, because of its small size, the whole complex may have been emplaced quite rapidly, so that the whole igneous sequence was formed during the same reversed interval of the geomagnetic field. Thus there is a marked contrast with the central intrusive complexes of Skye, Mull and Arran. The former with four centres was emplaced over at least three polarity intervals (Dagley & Mussett 1980), whereas Mull with three centres was emplaced over two polarity intervals (Mussett *et al.* 1980; Dagley *et al.* 1983) with a third polarity interval being recognized in the reversely magnetized, NW-trending basic dykes which cut the Loch Ba Felsite, the last major intrusion of the central intrusive complex. In Arran a polarity succession of at least R–N–R has been demonstrated by two stratigraphic sequences (Hodgson 1981).

Ardnamurchan is more like Rhum in being a small compact complex. Although Rhum has one of the largest gravity anomalies of all the centres the basic cylinder is estimated to be only 8–10 km in diameter (McQuillan & Tuson 1963). Rhum, too, is entirely reversely magnetized except for a few late dykes (Dagley & Mussett 1981). The overall mean directions for the two centres lie within 5° of each other and are the same at the 95 per cent confidence level (see Table 2). We think that the similarity is not fortuitous because the two centres lie along the same regional linear dyke swarm (Speight *et al.* 1982) which probably controlled the duration of igneous activity in that the mafic cylinders which underlie the central complexes are merely the higher-rising components of a more fundamental structure, namely the proposed extensive magma ridges from which the regional dyke swarms arose.

Radiometric dates using the K-Ar method have been obtained for several of the

Ardnamurchan intrusions. Mitchell & Reen (1973) found that Cone Sheets of Centre 1 lay in the range 59.2 ± 0.9 to 63.0 ± 4.2 Ma and Green & Wright (1974) obtained a date of 61.8 ± 3.1 Ma for one of the doleritic cone sheets (data have been recalculated using the decay constants recommended by Steiger & Jäger 1977). Cone Sheets of Centre 2 were found to have ages between 57.7 ± 1.2 and 62.3 ± 2.0 Ma while Ring intrusions of Centre 3 lie between 57.7 ± 0.9 and 59.0 ± 0.8 Ma (Mitchell & Reen 1973). Miller & Brown (1965) reported a date of 56 ± 6 Ma for the youngest intrusion, the quartz-monzonite of Centre 3. Two K-Ar age determinations on biotites separated from this body average 60.4 ± 1.1 Ma (Beckinsale & Walsh, private communication). A Rb-Sr whole-rock isochron for gabbros, dolerites and acid rocks from Centre 3 gives an age of 60.0 ± 1.7 Ma with a MSWD of 1.8 (Beckinsale & Walsh, private communication).

These dates suggest that the Ardnamurchan activity took place at more or less the same time as that in Mull and Rhum and we suggest that the reversed interval recorded in Ardnamurchan is the same as that recorded in Rhum. Both the Ardnamurchan and Mull central complexes followed the same plateau basalt sequence and we suggest that the reversed interval recorded in Ardnamurchan (and Rhum) is the same as the first reversed interval recorded in Mull. Other evidence (Meighan *et al.* 1982) shows that the Rhum centre preceded the lavas and central intrusive complex of Skye. The Skye lavas (Wilson, Dagley & Ade-Hall 1972) and the early Basic centres (Dagley & Mussett 1980) are reversely magnetized and dates for the centre (Dickin 1981, unpublished work of this laboratory) show it was contemporary with both Ardnamurchan and Rhum. Thus it is most probable that both these centres were intruded during the first reversed interval recorded in Skye. On the time-scale of Harland *et al.* 1982) this common reversed interval recorded in Rhum and Ardnamurchan and by the lavas and early intrusions of Mull and Skye would be Chron 26r, which lasted from 61.00 to 58.19 Ma.

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